



Dynamics characteristic of pedestrians' particular overtaking behavior based on an improved social force model

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ABSTRACT

The overtaking behavior of individuals under anxious panic is a major causative factor of crowded trampling accidents. In order to simulate the overtaking behavior of individuals experiencing anxiety and panic in a single-exit room within a normal pedestrian flow, an improved social force model is proposed to simulate crowd movement at local medium and high density through introducing the bypass overtaking mechanism, the side overtaking mechanism, and the side avoidance mechanism generated by association. The numerical simulation verifies the model's validity. The effect of the size and number of overtaking individuals on the pedestrian flow are explored. The results show that the new model can simultaneously reproduce a variety of overtaking behaviors and their associated behaviors in line with reality. The larger size of the overtaking individuals are, the easier it is to produce overtaking behaviors and the more difficult it is to overtake. With the increase of the number of overtaking individuals, the evacuation time becomes longer, but there is no linear relationship between them. When the initial pedestrian density is 0.37 ped/m², with time evolution, the appearance frequency of local density greater than 0.8 ped/m² suddenly increases, and the comfort and safety of pedestrian movement decreases. The results of this paper can enrich the social force simulation model of pedestrian flow and provide a theoretical guidance for the evacuation of pedestrian flow with a few overtaking individuals in the room.

1. Introduction

Carrying out the modeling and simulation of pedestrian dynamics under particular behaviors (such as crowding, grabbing, etc.), exploring crowd behavior laws and control methods is of great practical significance for preventing crowded stampede accidents. The overtaking behavior under individual anxiety and panic is a typical particular behavior, and the local information generated by this behavior is transmitted to other pedestrians with different perception abilities and decision-making performances around them, making the interaction between pedestrians complex and affecting group behavior. Many scholars have studied the overtaking behavior of pedestrians [1–14]. When the overtaking behavior occurs at the bottleneck of the building (such as the exit of the room), its impact is further amplified, directly affecting pedestrian evacuation safety and the overall traffic capacity of the building. Performing the simulation research of pedestrian flow in the room can analyze the pedestrian behavior factors affecting the evacuation efficiency of the room and obtain the method to avoid pedestrian congestion. Meanwhile, a room is essentially an area with limited space, and models simulating pedestrian movement in the room can be applied to study other motion scenarios

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Table 1
Comparisons between this paper and the relevant researches in recent years.

Authors	Behaviors	Model	References
Gao et al.	Bypass overtaking	Circle	Ref. [44]
Mu	Bypass overtaking and side overtaking	Ellipse	Ref. [14]
Zhang et al.	Bypass avoidance	Three circles	Ref. [45]
Mu	Bypass avoidance and side avoidance	Ellipse	Ref. [14]
This paper	Bypass overtaking, side overtaking and the association produced by side avoidance	Circle (variable radius)	–

through appropriate extensions. As a continuous micro-simulation model, the social force model has been widely used in pedestrian flow simulation [15–38]. The original social force model [39] generally uses fluctuations to represent the behavioral heterogeneity of individuals in a crowd lacking a reasonable mechanism for active overtaking. In order to simulate real-life scenarios more realistically, it is necessary to improve the original social force model when simulating pedestrian flows containing active overtaking behavior.

Ding et al. [7] analyzed the formation mechanism of adverse effects of overtaking behavior, proposed three negative effect evaluation indicators of trajectory offset rate, velocity entropy and velocity relative range, and determined the weight coefficient of the evaluation index based on the entropy weight method and constructed an evaluation model. Yuen and Lee [8] constructed the overtaking force as a power index function according to the horizontal distance, vertical distance and surrounding crowd density between pedestrians and realized active overtaking through the overtaking force. The model considered the direction and intensity of overtaking. Jia et al. [9] considered the behavioral characteristics of changing the desired speed direction and maintaining the desired speed magnitude during actively overtaking and changed the driving force to a combined force of forward force, lane changing force and conflict force. Zhang et al. [10] improved the social force model by considering factors such as pedestrians' sensitivity to speed differences, preference for the shortest path and social traffic rules to determine the direction of overtaking. Ye et al. [11] studied the overtaking behavior of pedestrians on stairs at different service levels, and the results showed that the intensity of overtaking behavior increased rapidly with the density from low to medium and the amount of available movement space for pedestrians was the main factor causing the above phenomenon. Tang et al. [12] established a continuous overtaking behavior model based on the speed-time domain to avoid conflicts with other pedestrians by changing the moving speed and direction of pedestrians. Shi et al. [13] improved the traditional social force model considering passenger overtaking behavior, and built a passenger walking model in the confluence channel of subway stations. It should be noted that some scholars have studied the overtaking behavior in vehicle traffic flow [40–43].

In summary, the current research on improving social force models considering overtaking behavior mainly focuses on bypass overtaking within channels, with less on overtaking behavior within rooms, and less on side overtaking behavior and associated behavior. In addition, the existing studies mainly consider a single overtaking behavior or avoidance behavior, but do not consider these behaviors at the same time, as shown in Table 1. In general, when pedestrians leave from a single-exit room, the local density of pedestrians near the exit will increase over time, and the interaction among pedestrians will become more complex. Pedestrians with urgent travel needs will not only make bypass overtaking but also perform side overtaking. Ordinary bypass overtaking will gradually evolve into “passing by” bypass overtaking, and will also be associated with side avoidance behavior [14]. How to describe these behaviors and how these factors affect the pedestrian dynamics within a room? This is an interesting but still open problem. Thus in this study, we investigate the overtaking behavior dynamic and modeling methods, and try to build a novel social force model to describe overtaking behavioral characteristics of individuals under anxious panic.

The paper is organized as follows. In Section 2, we propose an improved social force model considering overtaking behavior and side avoidance behavior by introducing the bypass overtaking mechanism, the side overtaking mechanism, and the side avoidance mechanism. In Section 3, numerical simulations are carried out and the results are discussed. Finally, we summarize our finding and discuss the future research in the area.

2. Model

2.1. Special overtaking behavior analysis

Fu et al. [46] believe that pedestrians will cause acquired panic due to inner anxiety, that is, anxiety panic. Helbing et al. [47] proposed that panic was a prerequisite for uncoordinated movement to occur. Social psychology believes that panicked pedestrians exhibit inappropriate and ruthless behavior, such as crowding and even life-threatening overcrowding. This article refers to overtaking behaviors caused by anxiety and panic, as special overtaking behaviors. Compared with previous overtaking behaviors, special overtaking behaviors have the following characteristics: Firstly, it occurs in local medium and high-density situations; Secondly, it contains two forms of overtaking: bypass overtaking and side overtaking; Thirdly, under certain conditions, there may be associated side avoidance behavior. In this article, pedestrians with special overtaking behaviors are called overtaking pedestrians; the rest are called obstructing pedestrians. When the overtaking space meets the side overtaking condition, overtaking pedestrians can independently complete bypass overtaking and side overtaking. When the overtaking space does not meet the side overtaking condition, overtaking pedestrians will issue a “request to give way signal”, and obstructing pedestrians on both sides of the target

movement gap will accept and respond to the signal with creating overtaking space for the overtaking pedestrian through side avoidance. Here, we divide special overtaking behavior into walking trajectory deviation stage, overtaking stage and the overtaking end stage.

Under normal circumstances, overtaking pedestrians will leave the room in the shortest time. We indirectly reflects the length of time it takes to leave the room using the desired speed realization proposed by Helbing when defining the level of pedestrian anxiety [48]. The larger desired speed realization, the shorter time it takes. When the local density of people in the room is low, the maximum desired speed is achieved, and the shortest time is required when leaving from the room along the shortest path. However, when the local density of pedestrians in the room is high, the time required to leave the room along the shortest path may not be the shortest, and it may also be affected by factors such as the speed of other pedestrians and the delay time caused by giving away. The improved model is suitable for local medium and high-density pedestrian flow, where the speed difference of obstructing pedestrians in the room is not significant and overtaking pedestrians will not exhibit following behavior. Therefore, the speed factor of obstructing pedestrians can be ignored when selecting the path for overtaking pedestrians. In addition, we assume that when a overtaking pedestrian is in the stage of trajectory deviation, a “request to give way signal” is issued, and obstructing pedestrians will immediately respond and begin to give way. So the delay time caused by obstructing pedestrians giving way does not affect the path selection of overtaking pedestrians. Based on the above considerations, overtaking pedestrian are assumed to leave the room in the shortest time, that is, the shortest path to leave the room.

2.2. Original social force model

The pedestrian flow social force model was first proposed by Helbing and Molnar in 1995 [39]. As shown in Eq. (1), the force experienced by pedestrian movement is divided into self-driving force $\vec{f}_i^0$, force between pedestrians $\vec{f}_{ij}$, and force between pedestrian and obstacle $\vec{f}_{iw}$.

$$m_i \frac{d\vec{v}_i}{dt} = \vec{f}_i^0 + \sum_{j(\neq i)} \vec{f}_{ij} + \sum_w \vec{f}_{iw} \quad (1)$$

Here, self-driving force $\vec{f}_i^0$ refers to the “social force” that is transformed from the subjective consciousness of pedestrians on their behavior, reflecting a willingness of pedestrians to achieve their desired speed, as shown in Eq. (2).

$$\vec{f}_i^0 = m_i \frac{v_i^0(t)\vec{e}_i^0 - \vec{v}_i(t)}{\tau_i} \quad (2)$$

In Eq. (2), m_i is the mass of the pedestrian; $v_i^0(t)$ is desired speed; $\vec{e}_i^0$ is the unit vector in the desired direction; $\vec{v}_i(t)$ is the walking speed for pedestrians; τ_i is relaxation time.

The interaction force $\vec{f}_{ij}$ between people mainly includes the social repulsive force $\vec{f}_{ij}^s$ generated by pedestrians trying to maintain a certain distance (generated when the distance between pedestrian center points is greater than the radius sum of two people) and the contact force $\vec{f}_{ij}^p$ between bodies (generated when the distance between pedestrian center points is less than the radius sum of two people). The contact force $\vec{f}_{ij}^p$ between the bodies includes the squeezing force $\vec{f}_{ij}^{p1}$ and friction force $\vec{f}_{ij}^{p2}$ between the bodies, as shown in Eq. (3) ~ Eq.(5).

$$\vec{f}_{ij} = \vec{f}_{ij}^s + \vec{f}_{ij}^p \quad (3)$$

$$\vec{f}_{ij}^s = A_i \exp\left(\frac{r_{ij} - d_{ij}}{B_i}\right) \vec{n}_{ij} [\lambda_i + (1 - \lambda_i) \frac{1 + \cos\beta_{ij}}{2}] \quad (4)$$

$$\vec{f}_{ij}^p = \vec{f}_{ij}^{p1} + \vec{f}_{ij}^{p2} = k_1 g(r_{ij} - d_{ij}) \vec{n}_{ij} + k_2 g(r_{ij} - d_{ij}) \Delta v_{ji}^t \vec{t}_{ij} \quad (5)$$

In Eqs. (4) and (5), A_i and B_i are constants, the radius sum of pedestrians i and j is $r_{ij} = r_i + r_j$, and the distance between pedestrians i and j is $d_{ij} = \|\vec{d}_i - \vec{d}_j\|$. In addition, $\vec{n}_{ij} = (n_{ij}^1, n_{ij}^2)$, $\Delta v_{ji}^t = (\vec{d}_i - \vec{d}_j) \cdot \vec{t}_{ij}$, $\cos\beta_{ij} = -\vec{n}_{ij} \cdot \vec{e}_i$, $\vec{t}_{ij} = (-n_{ij}^2, n_{ij}^1) \vec{e}_i$. λ_i is introduced to describe the effect of anisotropy in the field of view of pedestrians during movement, it is set $0 \leq \lambda_i \leq 1$. We assume $\lambda_i = 0.3$ which means that a pedestrian in front of the field of view will have a greater impact on the current pedestrian's motion decisions than a pedestrian behind them.

$$g(x) \text{ is a piecewise function } g(x) = \begin{cases} 0 & x < 0 \\ x & x \geq 0 \end{cases}.$$

The interaction force $\vec{f}_{iw}$ between people and obstacles mainly includes the repulsive force $\vec{f}_{iw}^s$ generated by pedestrians and obstacles trying to maintain a certain distance (generated when the distance between pedestrians and obstacles is greater than the radius of people) and the contact force $\vec{f}_{iw}^p$ between the body and obstacles (generated when the distance between pedestrians and obstacles is smaller than the radius of people). The contact force $\vec{f}_{iw}^p$ between the body and the obstacle includes the squeezing force $\vec{f}_{iw}^{p1}$ and friction force $\vec{f}_{iw}^{p2}$ between the body and the obstacle, as shown in Eq. (6) ~ Eq. (8).

$$\vec{f}_{iw} = \vec{f}_{iw}^s + \vec{f}_{iw}^p \quad (6)$$

$$\vec{f}_{iw}^s = A_w \exp\left(\frac{r_i - d_{iw}}{B_w}\right) \vec{n}_{iw} \quad (7)$$

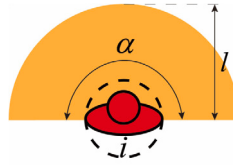


Fig. 1. Path search range for a overtaking pedestrian.

$$\vec{f}_{iw}^p = \vec{f}_{iw}^{p_1} + \vec{f}_{iw}^{p_2} = k_1 g(r_i - d_{iw}) \vec{n}_{iw} + k_2 g(r_i - d_{iw}) \Delta v_{wi}^t \vec{t}_{iw} \quad (8)$$

In Eq. (8), $\Delta v_{wi}^t = \vec{d}_i \cdot \vec{t}_{ij}$.

2.3. Improved social force model

The social force model has been widely used in the field of pedestrian flow simulation. However, the original social force model still has some problems in describing overtaking behavior due to the lack of a reasonable overtaking mechanism. In the original social force model, pedestrians rely on repulsive force to achieve bypass and overtaking, but the practical rationality of this overtaking phenomenon has limitations. Firstly, the speed of overtaking pedestrians must be significantly larger than the speed of the overtaken pedestrian, and when the speed difference is small, the overtaking process lasts for a long time, which is not in line with reality; Secondly, the two pedestrians must be within the search range with a noticeable repulsive effect. The repulsive force of the pedestrians decreases exponentially with the increase of the search radius. When the search radius is between 0 m to 0.2 m, the repulsive force is about 164 N to 573 N (These values are calculated based on the original social force model), and the effect of the force is very obvious. When the search radius is 0.4 m [39], the repulsive force is about 13 N (This value is calculated based on the original social force model), the effect of the force is not apparent; Thirdly, assuming that the repulsion is removed [10], the overlapping of the two pedestrian models during the overtaking process is relatively small. In addition, the original social force model cannot simulate the side overtaking behavior of pedestrians. When the distance between pedestrians is less than 0.2 m during side overtaking, the self-driving force of the side overtaking pedestrian is not enough to overcome the combined repulsive force generated by the two overtaken pedestrians, and the side overtaking pedestrian can only follow the target's movement gap and cannot complete the overtaking. Most of the existing improved social force models study overtaking behavior with low initial density in the channel, which is insufficient to simulate the “passing by” bypass overtaking behavior, side overtaking behavior, and associated side avoidance behavior in local medium to high-density situations in the room. Therefore, it is necessary to improve the original social force model to simulate overtaking behavior in the room.

By redefining the desired direction of self-driving force in the original social force model, the movement patterns of overtaking pedestrians during overtaking are reflected. The possibility of simulating side overtaking behavior is realized by narrowing the search range of repulsive forces in the original social force model. Simulating the switching between the front and side of a pedestrian can be realized by changing the radius of the pedestrian model. The effect of the “request to give way signal” is reflected by constructing a new mechanical model. As shown in Fig. 1, we consider the path search range when a overtaking pedestrian overtakes and design two related parameters: search radius l and search angle α .

$$\gamma_{ij,1} = \begin{cases} 1 & p = 1 \\ 0 & p = 0 \end{cases} \quad (9)$$

$$\gamma_{ij,2} = \begin{cases} 1 & d_{ij} \leq l_i \\ 0 & d_{ij} > l_i \end{cases} \quad (10)$$

$$\gamma_{ij,3} = \begin{cases} 1 & \frac{\vec{e}_i^0 \cdot \vec{d}_{ij}}{\|\vec{e}_i^0\| \|\vec{d}_{ij}\|} > 0 \\ 0 & \text{others} \end{cases} \quad (11)$$

Here, $\gamma_{ij,1}$ is pedestrian type identification parameter, $p = 1$ represents the overtaking pedestrian, $p = 0$ represents the obstructing pedestrian. $\gamma_{ij,2}$ reflects the search radius of pedestrian side overtaking behavior, d_{ij} is the distance between pedestrians i and pedestrians j , and l_i is the search radius for side overtaking behavior. $\gamma_{ij,3}$ is the parameter reflecting the search angle of pedestrian's side overtaking behavior, $\vec{e}_i^0$ is the current speed direction of the pedestrian, $\vec{d}_{ij}$ is the vector from pedestrian i pointing to the pedestrian j . When the angle between $\vec{e}_i^0$ and $\vec{d}_{ij}$ is less than 90 degrees, pedestrian j is within the search angle of side overtaking behavior for pedestrian i .

2.3.1. Overtaking behavior

The walking trajectory deviation [49] refers to the process in which pedestrians move from their initial position to the target movement gap during the overtaking. Overtaking pedestrians mainly complete the selection of overtaken pedestrians, the selection

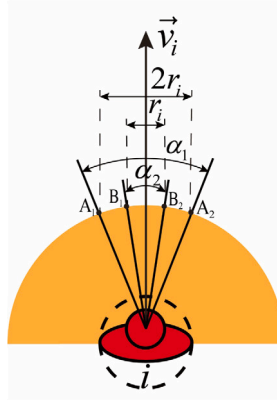


Fig. 2. Schematic diagram of strategy selection for overtaking direction.

of overtaking direction, and the transformation of desired direction during the trajectory deviation stage. During the overtaking process, the horizontal distance between the particular pedestrian and the overtaken pedestrian remains basically unchanged, while the longitudinal distance gradually decreases to 0 and then increases in the opposite direction [49]. After the overtaking, the desired direction of the overtaking pedestrian points to the exit and he/she continues walking towards the exit.

The overtaking direction includes three types: straight ahead, left front, and right front. As shown in Fig. 2, A_1 and A_2 are the intersection points of two straight lines with a distance $2r_i$ in front of the overtaking pedestrian and the boundary of the search area, respectively; B_1 and B_2 are the intersection points of two straight lines with a distance r_i in front of the overtaking pedestrian and the boundary of the search area, respectively; α_1 represents the angle at which A_1 and A_2 are connected to the center of the pedestrian model; α_2 represents the angle at which B_1 and B_2 are connected to the center of the pedestrian model. When there are no pedestrians in the search angle α_1 , overtaking pedestrians can overtake the pedestrians in front of them from the front. When there are pedestrians in the search angle α_1 and no pedestrians in the search angle α_2 , overtaking pedestrians can overtake the pedestrians in front of them from the side. When there are pedestrians within the search angle α_2 , overtaking pedestrians can overtake the pedestrian ahead from the left or right front. Overtaking pedestrians choose the shortest path to leave the room due to anxiety and panic. The shortest path is a straight path when overtaking pedestrians can overtake directly ahead without the need for side movement. However, when the local density of pedestrians in the room is high and overtaking from the front is blocked, the overtaking pedestrian needs to move to the left or right front to complete the overtaking. In this case, the shortest path must ensure the maximum desired speed achievement, that is, the minimum side distance.

Directional derivative is used to describe the strategy of selecting overtaken pedestrians for overtaking pedestrians. Overtaking pedestrians select pedestrians with larger directional derivatives within the search range to overtake, and the formula is as follows:

$$\left. \frac{\partial f}{\partial p} \right|_{(x_i, y_i)} = \nabla f(x_i, y_i) \cdot \vec{e}_p = \|\nabla f(x_i, y_i)\| \cos \phi \quad (12)$$

Here, $\nabla f(x_i, y_i)$ represents the gradient direction vector when the overtaking pedestrian walks, $\vec{e}_{p1}$ and $\vec{e}_{p2}$ (collectively referred to as $\vec{e}_p$) represent the unit vectors of the connection direction between the overtaking pedestrian and the pedestrians (except the pedestrian directly in front of it) within the search range, and ϕ is the angle between $\nabla f(x_i, y_i)$ and $\vec{e}_p$. As shown in Fig. 3, ϕ_1 represents the angle between $\nabla f(x_i, y_i)$ and $\vec{e}_{p1}$, and ϕ_2 represents the angle between $\nabla f(x_i, y_i)$ and $\vec{e}_{p2}$.

We use directional derivatives to describe the selection of side displacement direction for overtaking pedestrians when they overtake pedestrians ahead. Overtaking pedestrians move sidely along the direction with a larger directional derivative. The formula is as follows:

$$\left. \frac{\partial f}{\partial d} \right|_{(x_i, y_i)} = \nabla f(x_i, y_i) \cdot \vec{e}_d = \|\nabla f(x_i, y_i)\| \cos \theta \quad (13)$$

Here, $\vec{e}_{d1}$ and $\vec{e}_{d2}$ (collectively referred to as $\vec{e}_d$) represent the unit vectors of the tangent direction between the overtaking pedestrian and the pedestrian directly in front of it within the search range, and θ is the angle between $\nabla f(x_i, y_i)$ and $\vec{e}_d$. As shown in Fig. 3, θ_1 represents the angle between $\nabla f(x_i, y_i)$ and $\vec{e}_{d1}$, and θ_2 represents the angle between $\nabla f(x_i, y_i)$ and $\vec{e}_{d2}$.

We redefine the desired direction of self-driving force in the original social force model to reflect the motion law of overtaking pedestrians during overtaking. Fig. 4 shows desired direction change for overtaking pedestrians during bypass and overtaking. During the deviation stage of the walking trajectory, the desired direction $v_i^0 \vec{e}_i^0$ of the overtaking pedestrian is the vertical direction of the line connecting the overtaking pedestrian and the overtaken pedestrian, as shown in Fig. 4(a). During the overtaking stage and the overtaking end stage, the desired direction $v_i^0 \vec{e}_i^0$ of the overtaking pedestrian points towards the exit, as shown in Fig. 4(b).

Fig. 5 shows desired direction change for overtaking pedestrians during side overtaking. During the deviation stage of the walking trajectory, the desired direction $v_i^0 \vec{e}_i^0$ of the overtaking pedestrian is the vertical direction of the line connecting the overtaking

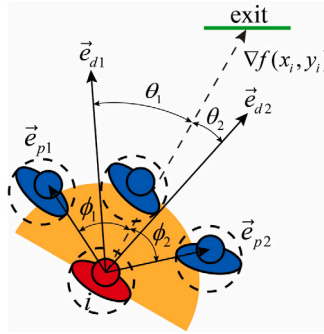


Fig. 3. Schematic diagram of strategy selection for overtaken pedestrians.

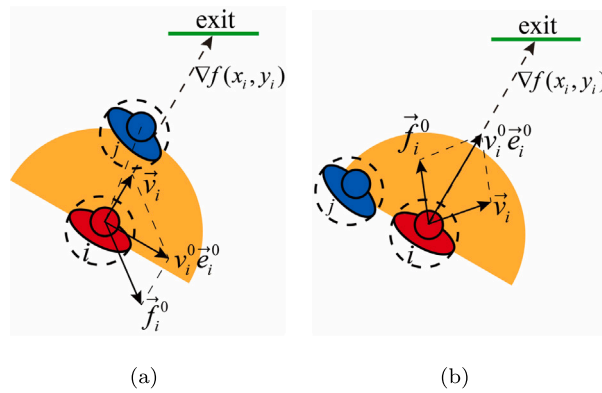


Fig. 4. Schematic diagram of desired direction change for overtaking pedestrians during bypass overtaking. (a) the deviation stage of the walking trajectory; (b) the overtaking stage and the end of the overtaking stage.

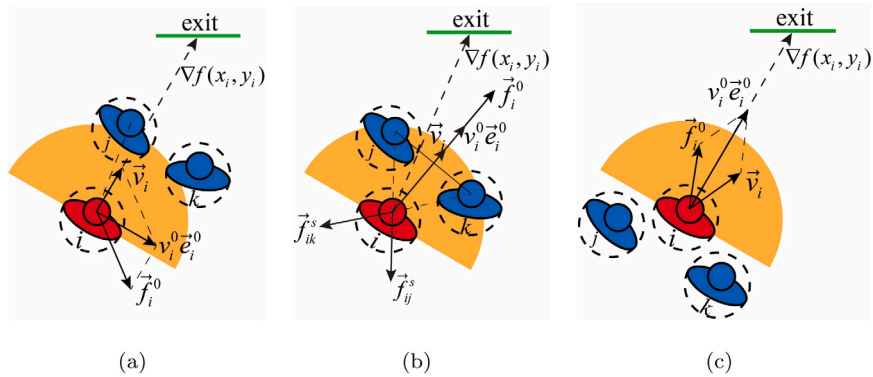


Fig. 5. Schematic diagram of desired direction change for overtaking pedestrians during side overtaking. (a) the deviation stage of the walking trajectory; (b) the overtaking stage; (c) the end of the overtaking stage.

pedestrian and the overtaken pedestrian, as shown in Fig. 5(a). During the overtaking stage, the desired direction $v_i^0 \vec{e}_i^0$ of the overtaking pedestrian points towards the center of the target movement gap, as shown in Fig. 5(b). And at the end of the overtaking stage, the desired direction $v_i^0 \vec{e}_i^0$ of the overtaking pedestrian points towards the exit, as shown in Fig. 5(c).

Pedestrians in different positions within the room have different desired directions towards the exit, as shown in Fig. 6. When pedestrians are located on both sides of the exit area, the desired directions $\vec{n}_{ie1}$ and $\vec{n}_{je1}$ point towards the central position of the exit. When pedestrians are located in the exit area, the desired directions $\vec{n}_{ie2}$ and $\vec{n}_{je2}$ point towards a position perpendicular to the exit.

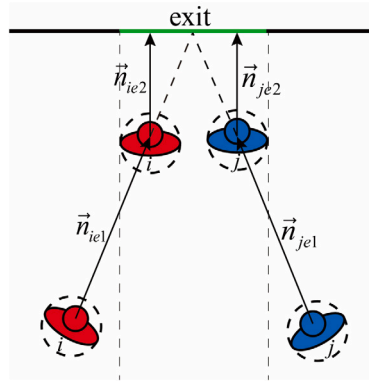


Fig. 6. Schematic diagram of the desired direction towards the exit with position.

The desired direction of the pedestrian is represented by $\vec{n}$, as shown in Eq. (14).

$$\vec{n} = \begin{cases} \vec{n}_E & \gamma_{ij,1}\gamma_{ij,2}\gamma_{ij,3} = 1 \\ \vec{e}_i^0 & \gamma_{ij,1}\gamma_{ij,2}\gamma_{ij,3} = 0 \end{cases} \quad (14)$$

Here, $\gamma_{ij,1}$ is pedestrian type identification parameter, $\gamma_{ij,2}$ is a parameter reflecting the search radius of pedestrian side overtaking behavior, $\gamma_{ij,3}$ is a parameter reflecting the search angle of pedestrian side overtaking behavior parameters; $\vec{n}$ is the desired velocity direction vector for pedestrians.

When a overtaking pedestrian i is located in the center of the target movement gap, the desired direction of the overtaking pedestrian $\vec{n}_E$ points towards the center of the target movement gap. That is, when the side distance is $l_{ik} = l_{ij}$, the desired direction of the overtaking pedestrian $\vec{n}_E$ is perpendicular to the direction of the line connecting pedestrian j and pedestrian k , and the perpendicular foot is the midpoint of the line, as shown in Eq. (15). When a overtaking pedestrian i is not located in the center of the target movement gap, that is, when the side distance is $l_{ik} \neq l_{ij}$, the desired direction of the overtaking pedestrian $\vec{n}_E$ is the product of the line connecting pedestrian i and pedestrian j and the rotation matrix C , as shown in Eq. (15).

$$\vec{n}_E = \begin{cases} \vec{n}_{ij} + \vec{n}_{ik} & l_{ik} = l_{ij} \\ \vec{n}_{ij} \cdot C & l_{ik} \neq l_{ij} \end{cases} \quad (15)$$

When a overtaking pedestrian i is not located in the center of the target movement gap, he/she chooses movement gaps with larger directional derivatives for overtaking. The rotation matrix C is defined as follows: after selecting the left (right) side movement gap, the desired speed direction of the overtaking pedestrian points towards the line connecting the overtaking pedestrian and the target movement gap on the left (right) side, and rotates counterclockwise (clockwise) by 90 degrees, as shown in Eq. (16).

$$C = \begin{cases} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} & \left. \frac{\partial f}{\partial l_{dl}} \right|_{(x_i, y_i)} > \left. \frac{\partial f}{\partial l_{dr}} \right|_{(x_i, y_i)} \\ \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} & \left. \frac{\partial f}{\partial l_{dl}} \right|_{(x_i, y_i)} \leq \left. \frac{\partial f}{\partial l_{dr}} \right|_{(x_i, y_i)} \end{cases} \quad (16)$$

Eq. (17) shows an improved expression of the desired direction of the pedestrian.

$$\vec{f}_i^0 = m_i \frac{v_i^0(t)\vec{n} - \vec{v}_i(t)}{\tau_i} \quad (17)$$

The search range for repulsive force in the original social force model is relatively large, which is not enough to simulate the “passing by” bypass overtaking and side overtaking. In addition, the change rate of the repulsive force intensity in the original social force model increases significantly with the decrease of pedestrian spacing. Therefore, when there is a significant difference in pedestrian speed, overlapping rebound phenomenon is prone to occur. Previous studies mainly overcome this phenomenon by reducing the simulation step size. However, it will greatly reduce simulation efficiency. In summary, if the search range for repulsive force under the same intensity and the change rate of repulsive force can be reduced, and the intensity of the action can be adjusted to an appropriate size, the “passing by” bypass and side overtaking can be simulated without overlapping rebound. As shown in Fig. 7, curves I and II show the relationship between the repulsive force $\|\vec{f}_{ij}^s\|$ and the distance d_{ij} between pedestrians when the action intensity is 1 and A_i , respectively. In order to reduce the repulsive force change rate, the improved scheme in this paper is to intercept the section of the original repulsive force change curve (curve I) with a small change rate (red dashed line) as the new repulsive force change curve (red solid line). Then the repulsive force action curve (solid green line) is obtained by adjusting the repulsive force action intensity to ensure that there is no overlap and rebound between pedestrians. Comparing the green curve and

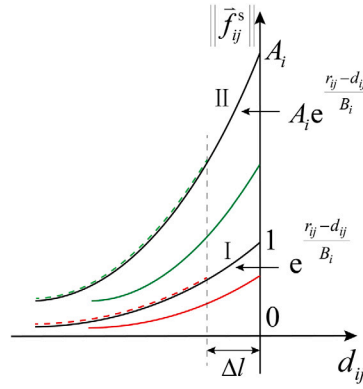


Fig. 7. Principle diagram of improving the effect of repulsive force.

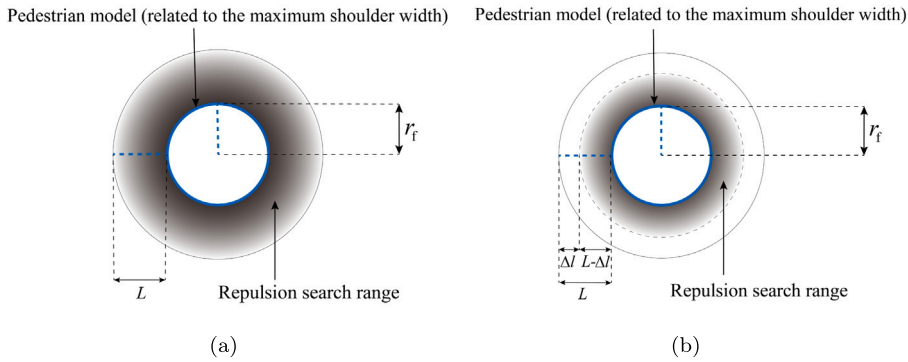


Fig. 8. Schematic diagram of repulsive force effect between pedestrians at the same action strength. (a) Repulsive force effect of original social force model; (b) Repulsive force effect of the improved social force model.

curve II, it can be seen that the change rate of repulsion force and the maximum strength of repulsion force between pedestrians are reduced after the improvement.

Fig. 8 shows the schematic diagram of the effect of repulsive force between pedestrians under the same action intensity. Fig. 8(a) shows the effect of repulsive force in the original social force model. Fig. 8(b) shows the effect of repulsive force in the improved social force model. The search scope of the repulsion force is the area between the inner boundary (blue circle) and the outer boundary (the solid black circle in Fig. 8(a), the dashed black circle in Fig. 8(b)). The size of the repulsive force from the inner boundary to the outer boundary decreases exponentially (illustrated by the gradual change of color in Fig. 8). In Fig. 8(b), the search range of repulsion force is changed from L to $L - \Delta l$, and the change rate of repulsion force near blue circle is reduced. In this way, the side overtaking behavior of pedestrian can be simulated. Otherwise, unrealistic rebound phenomenon will occur when the distance between pedestrians is small.

The side behavior of pedestrians only involves rotating at a certain angle in place, and almost no displacement, so the change of desired direction is not considered during rotating side. In previous studies, in order to describe the side behavior of pedestrians, pedestrians were abstracted as elliptical models, and moments were applied at the two focal points to achieve the rotation of the ellipse. In this article, we do not focus on the specific behavior of pedestrians when turning side, so pedestrians are abstracted into as circles. As shown in Fig. 9, when describing pedestrian's side overtaking and avoidance, the pedestrian is abstracted as a circle with a radius r_s . When describing the pedestrian walking forward, the pedestrian is abstracted as a circle with a radius r_f . The ratio r_f to r_s is set according to the ratio of maximum shoulder width to chest thickness in Chinese body size statistics.

2.3.2. Side avoidance behavior

The perception of environmental information in the original social force model includes two ways: the change in physical contact force caused by tactile signals and the change of social force caused by visual signals. In actual scenarios, sound signals can also cause changes in social forces, thereby affecting pedestrian's behaviors. The influence range of the "request to give way" signal issued by overtaking pedestrians is related to factors such as the strength of the sound source signal itself, the level of environmental noise, the degree of pedestrian attraction. Whether information stimulation will cause changes in individual behavior depends on the decision-making process after receiving the information. This article only focuses on pedestrians who receive and execute the signal, so the influence range of the signal itself is not considered.

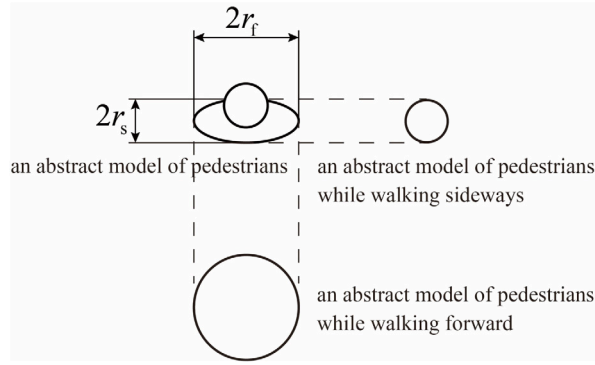


Fig. 9. Frontal and side models.

After the overtaking pedestrian sends a “request to give way” signal, pedestrians on both sides of the target movement gap will perform side avoidance, which includes six processes: deceleration, self-stopping, turning, waiting, turning again and starting. During the process of deceleration and self-stopping, pedestrians are induced by sound signal. The turning, waiting, and re-turning are described by turning rules. The starting process indicates that the pedestrians have completed the side avoidance behavior and resumed their movement. This article stipulates that only pedestrians on both sides of the target movement gap will be induced by the sound signal, as shown in Eq. (18).

$$\eta_s = \begin{cases} 1 & k = 1 \\ 0 & k = 0 \end{cases} \quad (18)$$

Here, η_s is a parameter describing for the influence ability to sound signals, $k = 1$ indicating that the pedestrian is about to make side avoidance, $k = 0$ indicating that the pedestrian will not perform side avoidance.

When the overtaking pedestrian is in the stage of walking trajectory deviation, pedestrians on both sides of the target movement gap will simultaneously complete deceleration, self-stopping and turning actions. The induction force of sound signal is similar to the attraction effect in the original social force model, but its direction of action is opposite. Therefore, the dimension of attraction in the original social force model can be used to construct the induction force sound signals $\vec{f}_{att}$, as shown in Eq. (19).

$$\vec{f}_{att} = \sigma A_i \exp\left(\frac{r_{ij} - d_{ij}}{B_i}\right) \eta_s (-\vec{e}_i^0) \quad (19)$$

Here, $A_i \exp\left(\frac{r_{ij} - d_{ij}}{B_i}\right)$ is the dimension of $\vec{f}_{att}$, η_s is a parameter describing for the influence ability to sound signals, $\sigma \in [0, 1]$ is the strength parameter.

Add $\vec{f}_{att}$ to the original social force model to complete the improvement of the social force model, as shown in Eq. (20).

$$m_i \frac{d\vec{f}_{ij}}{dt} = \vec{f}_{ij}^0 + \sum_{j(\neq i)} \vec{f}_{ij} + \sum_w \vec{f}_{iw} + \vec{f}_{att} \quad (20)$$

3. Simulation results

The parameter values in the simulation experiment are shown in Table 2. In order to eliminate the influence of randomness in the initial state of pedestrians on the results, the simulation results were averaged by 20 repeated simulation data. Firstly, we verify the effectiveness of this model.

In 2000, Helbing et al. [47] studied the impact of factors such as panic and impatience on evacuation and found that emergency exits were often ignored or not fully utilized under panic conditions. Therefore, we assume that overtaking pedestrians have a smaller path search range due to their inner anxiety and high local pedestrian density, as a result, the path search range is limited to the vicinity of their current location. Here, we set the search radius $l = 3r_f$ and search angle $\alpha = 180^\circ$. According to the Chinese body size statistics, the ratio of maximum shoulder width to chest thickness is taken as 2.

3.1. Model reliability verification

Fig. 10 gives the spatiotemporal evolution patterns of pedestrians at different time steps during the evacuation process with an initial density $\rho = 0.37$ ped/m² and $r_f = 0.3$ m. It can be found that the new model can reproduce the classical “arching phenomenon”.

Here, we verify the model’s validity through reproducing typical scene. Fig. 11 gives the dynamic spatiotemporal evolution patterns of the behavior of special and obstructing pedestrians during side overtaking. From Fig. 11(a), it can be seen that when

Table 2
Parameter values in the social force model.

Parameter	Meaning	Value [50]
v_i^0	Pedestrian desired speed	1 m/s
m_i	Pedestrian quality	80 kg
τ_i	Delay time	0.1 s
t	Simulation step size	0.1 s
v_o^0	Desired speed for obstructing pedestrian	1 m/s
v_s^0	Desired speed for overtaking pedestrian	2.5 m/s
r	Model radius for obstructing pedestrian (walking forward)	0.3 m
r'	Model radius for obstructing pedestrian (walking side)	0.15 m
r_f	Model radius for overtaking pedestrian (walking forward)	[0.26,0.36] m
r_s	Model radius for overtaking pedestrian (walking side)	[0.13,0.18] m
l	Search radius for overtaking pedestrian	$3r_f$
α	Search angle for overtaking pedestrian	180
λ_1	Anisotropy in the field of view	0.3
k_1	Squeezing force parameters	120000 kg/m ²
k_2	Friction force parameters	40000 kg/(m s)
A_w	Repulsive force strength (pedestrian-wall)	2000 N
B_w	Characteristic distance of repulsive force (pedestrian-wall)	0.08 m
A_i	Repulsive force strength (pedestrian-pedestrian)	2000 N
B_i	Characteristic distance of repulsive force (pedestrian-pedestrian)	0.04 m
ρ	Pedestrian initial density	[0.13,0.37] ped/m ²
L	Room length	20 m
W	Room width	20 m
D	Exit width	3 m

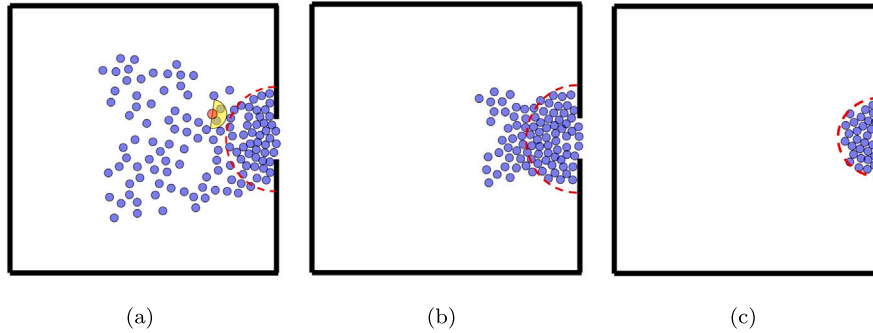


Fig. 10. The spatiotemporal evolution patterns during pedestrian evacuation as $\rho = 0.37$ ped/m² and $r_f = 0.3$ m. (a) $t = 10$ s; (b) $t = 20$ s; (c) $t = 30$ s.

obstructing pedestrians enter the search range of overtaking pedestrians, they slow down to 0 to make avoidance side. From Fig. 11(b), it can be seen that there are obstructing pedestrians in the direction of the line connecting the overtaking pedestrian and the exit center. The overtaking pedestrian first enters the walking trajectory deviation stage, moves to the center of the target movement gap, and then overtakes side, while obstructing pedestrians perform avoidance side. As seen from Fig. 11(c), after the overtaking is completed, both overtaking pedestrians and obstructing pedestrians transition to a forward walking state. From Fig. 11(d), (e), and (f), when a overtaking pedestrian walks near the exit, the desired direction changes again in order to leave the room faster, which is in accordance with reality. The behavior of overtaking pedestrians and obstructing pedestrians during these processes is consistent with the pedestrian behavior mentioned in the previous improvement of the social force model, which verifies the effectiveness of this model [47].

3.2. Size effect of overtaking pedestrians

In real scenarios, the size of pedestrians varies greatly with factors such as age, gender, weight and whether luggage is carried. The size of overtaking pedestrians has a significant impact on the choice of overtaking mode, so it is necessary to consider the factor of overtaking pedestrian size. Referring to the Chinese national standard GB/T 10000-1988, the range and corresponding relationships between maximum shoulder width and chest thickness for overtaking pedestrians are shown in Table 3. Accordingly, the radius of the overtaking pedestrian model is set from left to right to 0.26 m, 0.28 m, 0.3 m, 0.32 m, 0.34 m and 0.36 m, respectively. At the initial moment, 50, 75, 100, 125, and 150 people were randomly distributed in the room, corresponding to the initial densities of approximately 0.13 ped/m², 0.19 ped/m², 0.25 ped/m², 0.31 ped/m² and 0.39 ped/m², respectively. The overtaking pedestrian is initially located at coordinates $(-8, 6)$.

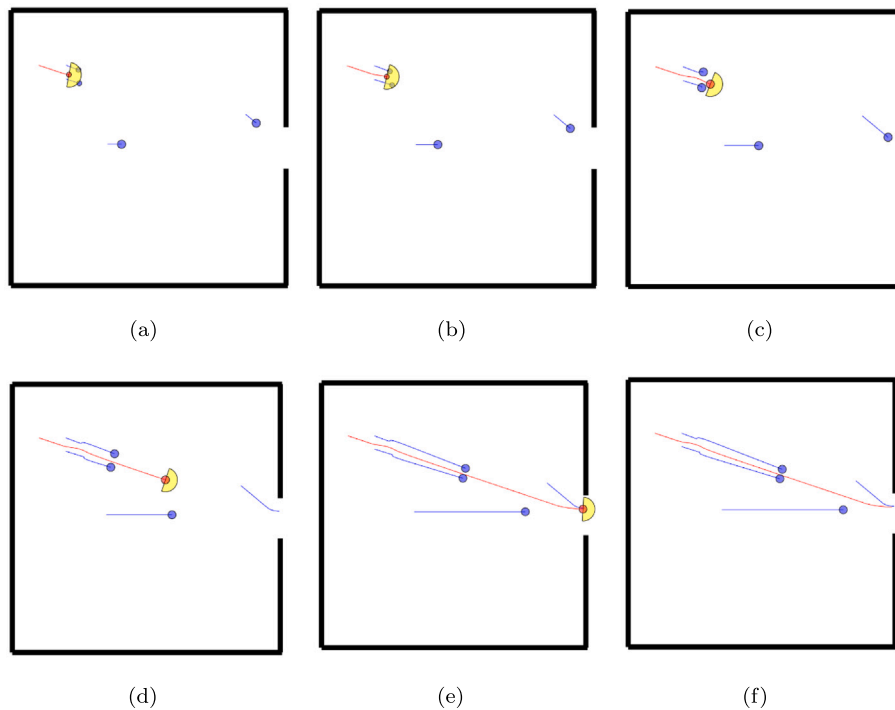


Fig. 11. Dynamic spatiotemporal evolution pattern of pedestrian behavior during side overtaking. (a) $t = 1.3$ s; (b) $t = 1.8$ s; (c) $t = 2.8$ s; (d) $t = 5.2$ s; (e) $t = 8.6$ s; (f) $t = 8.7$ s.

Table 3
Overtaking pedestrian's body size.

Item	Value					
Maximum shoulder(mm)	340	370	400	430	460	490
Chest thickness width(mm)	145	170	195	220	245	270

Fig. 12 shows the relationship between the time it takes for a overtaking pedestrian to leave the room and the radius of the overtaking pedestrian model under different initial densities. It can be seen that when the initial density is greater than 0.19 ped/m^2 , with the increase of the initial density at equal intervals, the time it takes for overtaking pedestrians to leave the room increases slowly and then quickly. This is because as the initial density increases, the influence of the initial position distribution of pedestrians on the behavior of overtaking pedestrians gradually weakens, and the local density near overtaking pedestrians gradually increases, which has an increasing impact on their behavior. For the same initial density, as the radius of the overtaking pedestrian model increases, the positive correlation between the time taken to leave the room and the overtaking pedestrian's size increases, indicating that the larger the overtaking pedestrian's size, the more likely and difficult to overtake. When the initial density is 0.13 ped/m^2 and the radius of the overtaking pedestrian model is less than 0.32 m , the relationship between the time taken by the overtaking pedestrian to leave the room and the size of the overtaking pedestrian fluctuates greatly. This is because when the initial density is small, the behavior of the overtaking pedestrian is greatly affected by the initial position distribution of the pedestrians. However, when the radius of the overtaking pedestrian model is greater than 0.34 m , the time it takes for the overtaking pedestrian to leave the room increases significantly, because the larger the size of the overtaking pedestrian, the more likely they are hindered by pedestrians forward, and the more likely they want to overtake, thereby prolonging their time. Therefore, in the case of medium and high density, it is necessary to monitor in real time whether there are large overtaking pedestrians in the room and take certain control measures at all times to avoid crowding and stampede accidents.

Fig. 13 shows the spatiotemporal evolution pattern of pedestrian evacuation with an initial density of 0.13 ped/m^2 and $t = 8 \text{ s}$ under the different radius of the overtaking pedestrian model. As can be seen from Fig. 13(a), (b), (c) and (d), the overtaking pedestrian is closer to the exit position, there is less overtaking behavior during subsequent walking, indicating that the difference in time required to leave the room is relatively small. In Fig. 13(e), the overtaking pedestrian is far away from the exit position, and there are more overtaking behaviors during subsequent walking, which can predict a longer time to leave the room. From Fig. 13(f), it can be found that the overtaking pedestrian is closer to the exit position, and there are more overtaking behaviors during subsequent walking, which indicates a longer time to leave the room. Compared Fig. 13(f) with Fig. 13(c), it can be observed that the interaction between overtaking pedestrians in the early stage of this time point is more complex in Fig. 13(c), which indirectly indicates that when the size of overtaking pedestrians increases to a certain extent, it takes longer to overtake.

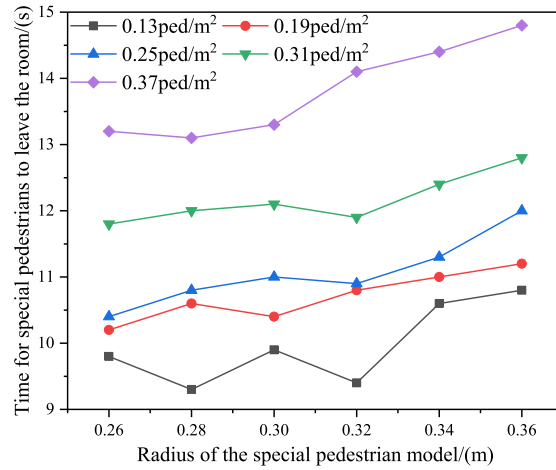


Fig. 12. Influence of overtaking pedestrian size on evacuation process under different initial densities.

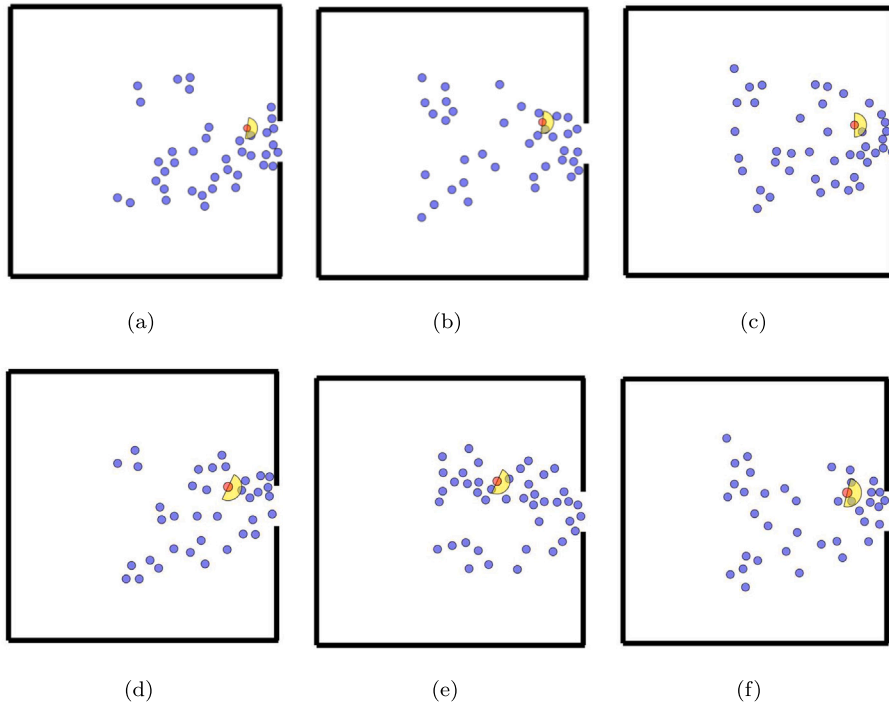


Fig. 13. Spatiotemporal evolution pattern of pedestrian evacuation as $\rho = 0.13 \text{ ped/m}^2$, $t = 8 \text{ s}$. (a) $r_f = 0.26 \text{ m}$; (b) $r_f = 0.28 \text{ m}$; (c) $r_f = 0.3 \text{ m}$; (d) $r_f = 0.32 \text{ m}$; (e) $r_f = 0.34 \text{ m}$; (f) $r_f = 0.36 \text{ m}$.

As pedestrians gather towards the exit, the local density of pedestrians in the room becomes uneven. Through the above analysis, it is found that the initial density indirectly affects the time it takes for overtaking pedestrians to leave the room by affecting the local density, so it is necessary to further analyze the change of local density during the process of overtaking pedestrians leaving the room. Local density is related to time and space. The local density near overtaking pedestrians has a significant influence on pedestrian behavior, so the area around overtaking pedestrians is taken as the local density statistical area, which is about 5 m^2 , as shown in Fig. 14.

Fig. 15 shows the statistical chart of the number of occurrences of local density values near overtaking pedestrians under different initial densities of pedestrians. It can be seen that when the initial density is low, the local density near overtaking pedestrians reaches medium-high density for a long time during the process of leaving the room. Moreover, as the initial density increases, the time when the local area is in a low-density state decreases, and the density value in a medium-high-density state increases.

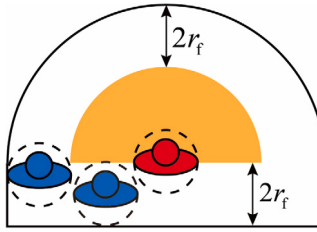


Fig. 14. Local density statistical area.

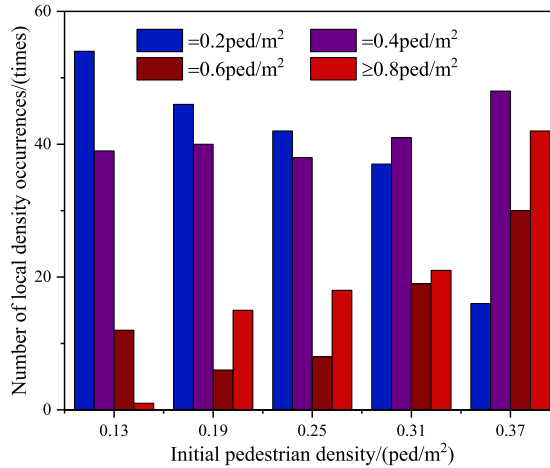


Fig. 15. Statistical chart of the number of occurrences of different local density values under different initial density of pedestrians.

Consequently, the higher the initial density, the more interaction between overtaking pedestrians and obstructing pedestrians during the process of leaving the room, and the relatively longer it takes to leave the room.

3.3. Effect of the number of overtaking pedestrians

Next, we explore how the number of overtaking pedestrians influence the evacuation. Here, the number of overtaking pedestrians is set 1,2,3,4 and 5 under different initial pedestrian densities. When there are 5 overtaking pedestrians at the initial moment, they are located at coordinates $(-8,6)$, $(-8,-6)$, $(-8,0)$, $(-6,6)$, and $(-6,-6)$, as shown in Fig. 16. When the remaining number of overtaking pedestrians is in the initial moment, the initial position is selected from left to right according to the above coordinates.

Fig. 17 shows the relationship between the time it takes for overtaking pedestrians to leave the room and the number of overtaking pedestrians under different initial densities. It can be seen that under the same initial density, the more overtaking pedestrians there are, the longer it takes to leave the room, but there is no linear relationship between them. This is because the presence of overtaking pedestrians makes the interaction between pedestrians more complex. With the increase of the number of overtaking pedestrians, those overtaking pedestrians who leave the room earlier will make more obstructing pedestrians gather at the exit, and the complexity of the interaction between overtaking pedestrians who leave later and obstructing pedestrians near the exit will further increase. When the number of overtaking pedestrians is constant, the time it takes for overtaking pedestrians to leave the room increases with the increase of the initial density. As shown in Fig. 15, as the initial density increases, the local density near the overtaking pedestrians also increases, and the behavior of overtaking pedestrians becomes more complex, resulting in a longer time to leave the room.

Fig. 18 shows the spatiotemporal evolution pattern of pedestrian evacuation with an initial density of 0.13 ped/m^2 and $t = 8 \text{ s}$ under different numbers of overtaking pedestrians. In Fig. 18(a), (b), and (c), overtaking pedestrians have all approached the exit, and it can be predicted that it will take approximately the same amount of time to leave the room. In Fig. 18(d) and (e), some overtaking pedestrians are still far away from the exit, so when they continue to move towards the exit, more obstructing pedestrians will gather at the exit, indicating that it will take longer for all overtaking pedestrians to leave the room.

4. Conclusions and outlook

In normal pedestrian flow, it is common for a small number of pedestrians to exhibit behaviors such as bypass due to emergencies. In order to study the overtaking behavior of individuals experiencing anxiety and panic during the evacuation of

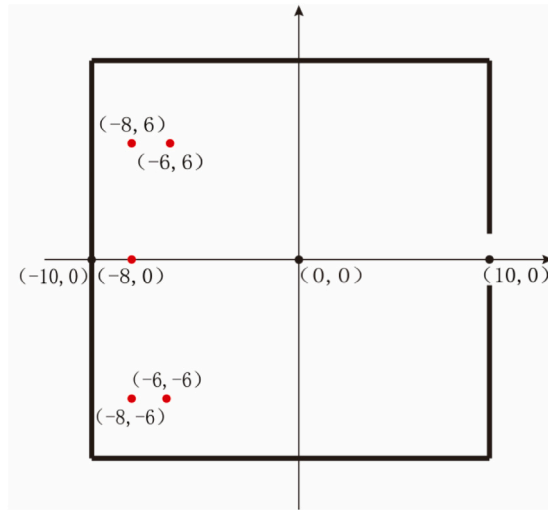


Fig. 16. Distribution of initial locations for 5 overtaking pedestrians.

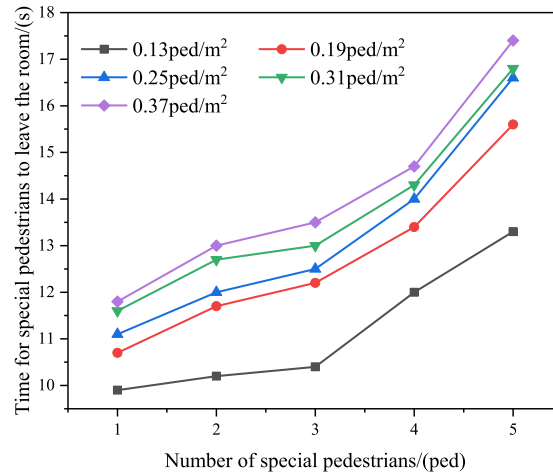


Fig. 17. The relationship between the time it takes for overtaking pedestrians to leave the room and the number of overtaking pedestrians under different initial densities.

single exit room within the normal pedestrian flow, an improved social force model is proposed, which solves the problem that the existing social force models cannot simultaneously simulate bypass overtaking behavior, side overtaking behavior, and side avoidance behavior generated by association. A new conversion mechanism between forward walking and side walking is proposed. Based on the original tactile and visual signals, auditory signals are introduced to make pedestrian behavior decision more realistic. The effect of repulsive force under the same repulsive force intensity is improved so that the improved model can simulate the “passing by” bypass overtaking behavior and side overtaking behavior. Moreover, a reasonable configuration of repulsive force intensity can achieve pedestrian collision avoidance to a certain extent, avoiding pedestrian collision avoidance only by reducing the simulation step size in previous research, and thus improving the simulation efficiency. The effectiveness of the model validity is verified by reproducing the “arching” and the overtaking phenomenon in real scenes. Then, the influence of the size and number of overtaking individuals on pedestrian flow is further explored. The results indicate that the larger the size of special individuals, the more likely they perform overtaking behavior and the more difficult for overtaking. With the increase of the overtaking pedestrians, the time it takes them to leave the room becomes long, but there is no linear relationship between them. In the case of medium and high density, it is necessary to monitor the size and number of pedestrians in the room and the local density around them in real time, and take certain control measures necessarily to improve the safety, comfort and rapidity of pedestrian evacuation and avoid crowding and stampede accidents.

In the future work, we will further explore path choice of special individuals’ behaviors (e.g., with luggage, blind people, riding a wheelchair) under anxious panic in more complicated environments such as with obstacles, diverse spatial building structures.

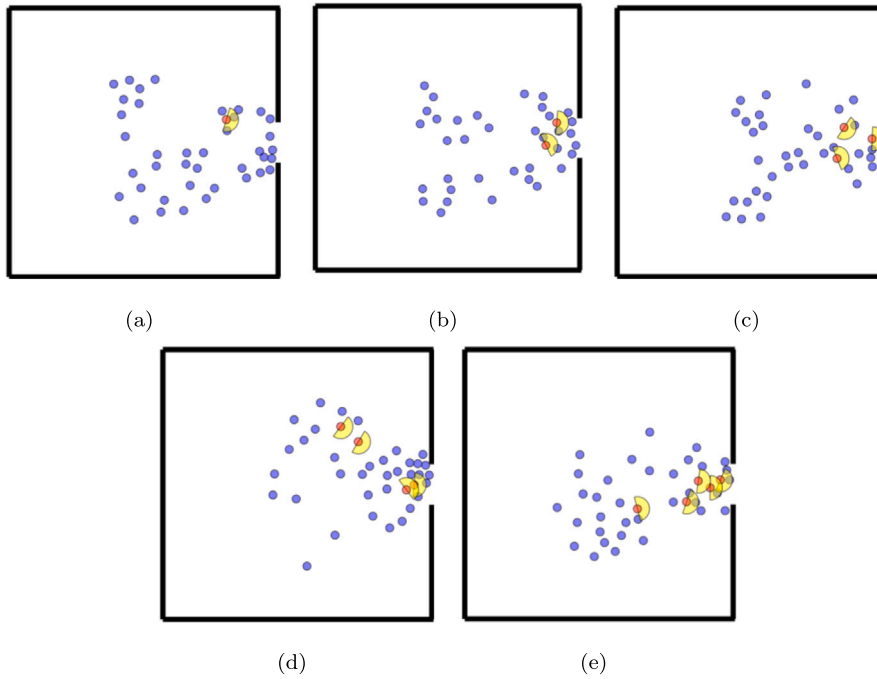


Fig. 18. The spatiotemporal evolution pattern of pedestrian evacuation with an initial density of 0.13 ped/m^2 and $t = 8 \text{ s}$ under different numbers of overtaking pedestrians. (a) 1; (b) 2; (c) 3; (d) 4; (e) 5.

CRediT authorship contribution statement

Jiangtao Tian: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xingli Li:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Qinghua Guo:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Hua Kuang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationship with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

The authors do not have permission to share data.

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